1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Case A - Hierarchical Path Triage Hypothesis: In men with localized prostate cancer treated by radical prostatectomy, PTEN loss is independently associated with shorter time-to-biochemical recurrence (biochemical recurrence defined as PSA ≥ 0.2 ng/mL post-surgery). This association is modulated by AKT activation (proxied by p-AKT) and by co-occurring alterations in AR signaling, mTOR pathway activity, FOXO and S6K signaling, as well as ERG (ERG rearrangement) and TP53 status. Importantly, the magnitude and direction of the PTEN loss effect differ by the mechanism of PTEN loss (PTEN copy-number loss vs PTEN mutation vs PTEN protein loss by IHC). The mechanistic substrate is that PTEN loss elevates PI3K/AKT/mTOR signaling, which cross-talks with AR signaling and downstream effectors (FOXO, S6K) to promote tumor progression and biochemical relapse; ERG and TP53 alterations may further modify this relationship. The study will test (1) the main effect of PTEN loss on time-to-biochemical recurrence, (2) effect modification by AKT activation, (3) differential effects by PTEN loss mechanism, and (4) interaction with co-alterations (AR signaling, mTOR, FOXO, S6K, ERG, TP53). If feasible, analyses will be conducted in Path A (integrated main model with modifiers) and Path B (stratified analyses by PTEN loss mechanism and AKT activation status) and will utilize AKT activation proxies (p-AKT) and pathway markers to refine mechanistic signal and clinical interpretability.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions